

Optimizing Input Representations for Word-in-Context: A Controlled Ablation Study

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Abstract. The Word-in-Context (WiC) task requires deciding whether a target word carries the same meaning in two different sentential contexts. When fine-tuning Transformer encoders for this task, a common and intuitive choice is to mark the target word with special tokens, on the assumption that such a signal helps the model locate the relevant span. We test this assumption through a controlled 2×2 factorial ablation on SuperGLUE WiC, separating the contribution of explicit target-span marking (span highlighting) from that of a lexical prefix that prepends the target word, and further decompose the prefix to isolate the role of the lemma from that of the part-of-speech (POS) tag. The design is evaluated on DeBERTa-large and RoBERTa-large using a 5-fold cross-validation on the union of training and development splits ($N = 6066$), with McNemar’s test, Holm–Bonferroni correction, and bootstrap confidence intervals. Span highlighting provides no benefit and, in our cross-validation, is associated with significantly lower out-of-fold accuracy on DeBERTa-large (removing the markers improves out-of-fold accuracy by 2.26 pp, $p_{\text{Holm}} < 0.001$); this outcome is consistent with prior observations in relation extraction that in-vocabulary markers can match or outperform newly introduced special tokens. The lexical prefix significantly improves both backbones when span markers are absent. We verify that the SuperGLUE WiC release exposes no usable POS field, so the prefix is effectively lexical; the decomposition therefore contrasts the target lemma against a constant control, and, especially on RoBERTa-large, the lemma is the active component. The contribution is a controlled, statistically grounded quantification of input-formatting choices that are frequently adopted without separate verification.

Keywords: Word-in-Context · Transformer Encoders · Span Highlighting · Ablation Study · Natural Language Processing

1 Introduction

Word-sense discrimination is a long-standing problem in computational linguistics: a word may take on different meanings depending on its context. This intuition is classically captured by the distributional view of meaning, summarised by Firth’s dictum that one shall know a word by the company it keeps [14]: the surrounding context, rather than the word form in isolation, determines the intended sense. The Word-in-Context (WiC) dataset operationalises this view

as a binary task: given two sentences and a shared target word, a model must decide whether the word carries the same meaning in both contexts [1]. WiC was subsequently incorporated into SuperGLUE [2].

With the rise of Transformer encoders such as BERT, RoBERTa, and DeBERTa [3–5], sentence-pair classification tasks are typically addressed through supervised fine-tuning. Beyond the choice of backbone, the way the input is represented can also influence performance. A natural choice for WiC is to mark the target word with special tokens, for example “Do you want to come over to my <TW>place</TW> later?”. The intuition is that explicit markers help the model focus on the relevant span. This strategy (span highlighting) is closely related to marking techniques used in relation extraction and other semantic tasks. Yet the special tokens are not part of the model’s original pre-training distribution and must be learned during fine-tuning; moreover, their effect can be conflated with other signals such as a prefix prepending the target word. This motivates the central question: does explicit span highlighting genuinely improve WiC, or does the useful signal mainly come from a lexical prefix?

To answer this, we pose three research questions. First, does explicit span highlighting improve WiC over an otherwise identical input, or is it unnecessary or even harmful (RQ1)? Second, does a lexical prefix improve WiC, and does it interact with span highlighting (RQ2)? Third, within the prefix, is the active component the target lemma or the part-of-speech tag (RQ3)?

We address these questions with a 2×2 factorial ablation that separates span marking from the lexical prefix, and further decompose the prefix into its lemma and part-of-speech components. Our study contributes a controlled ablation of target-span marking and a lexical prefix on SuperGLUE WiC, evaluated on DeBERTa-large and RoBERTa-large under a frozen training configuration with 5-fold cross-validation on $N = 6066$ examples, together with a statistical analysis based on McNemar’s test, Holm–Bonferroni correction, and bootstrap confidence intervals, and an error analysis of the headline effect.

Our main results are clear-cut. Span highlighting provides no benefit and, in our cross-validation, significantly lowers DeBERTa-large accuracy (a 2.26 pp drop, $p_{\text{Holm}} < 0.001$), whereas the lexical prefix significantly improves both backbones when markers are absent (up to +2.18 pp on RoBERTa-large). Decomposing the prefix shows that, on RoBERTa-large, the target lemma—rather than a content-free prefix token—carries the signal, and the error analysis indicates that the effect of the markers is diffuse across examples rather than concentrated on an identifiable subset.

The remainder of the paper is organised as follows. Section 2 reviews related work on WiC and on target marking in adjacent tasks. Section 3 describes the task, data, input conditions, models, and evaluation protocol. Section 4 reports the per-condition results, the statistical analysis of the contrasts, and the error analysis. Section 5 discusses the findings, Section 6 states the limitations, and Section 7 concludes.

2 Related Work

WiC was introduced by Pilehvar and Camacho-Collados as a benchmark for context-sensitive representations of word meaning [1], and included in SuperGLUE as a challenging test of contextual lexical discrimination [2]. A relevant caveat for any input-side study is that WiC does not test context and target word symmetrically: Liu et al. [16] probe lexical-semantic datasets with masked inputs and find that WiC-style tasks exhibit a strong context bias, with models able to exploit dataset regularities. This matters for our work, because it means that a lexical anchor on the target word is not trivially expected to help, and that any benefit deserves explicit measurement rather than assumption.

These tasks are typically addressed by fine-tuning Transformer encoders: BERT introduced bidirectional pre-training with masked language modeling [3], RoBERTa improved the pre-training procedure [4], and DeBERTa introduced disentangled attention between content and position [5]. In WiC these models are used as cross-encoders, where the two sentences are concatenated and the target word is not automatically privileged, which motivates explicit localization or lexical signals.

Marking the target span is a familiar device in adjacent tasks. In relation extraction, *Matching the Blanks* demonstrated that entity-centric representations based on marked spans are effective [8], and, more directly relevant to our question, Zhou and Chen [12] systematically compare marking variants and distinguish between markers that introduce *new special tokens* and markers that reuse *in-vocabulary punctuation*; the punctuation-based variant, which adds no out-of-distribution tokens, is a strong baseline. This distinction directly anticipates our finding for WiC: when a marker is realised through tokens that the model has never seen in pre-training, the benefit of marking is not guaranteed. MarkBERT similarly inserts in-vocabulary boundary markers and reports that marking can help or hurt depending on how the markers are realised [17]. Within WiC and adjacent tasks, XL-LEXEME highlights the target word in a bi-encoder for lexical semantic change [9], and participants in the MCL-WiC shared task used signals on the target word in multilingual settings [10, 11]; crucially, these systems *assume* highlighting to be beneficial and do not evaluate the contribution of the marking against a baseline without it. Our ablation addresses precisely this open question in the cross-encoder setting: the question is not whether to mark the target, but whether marking is necessary when compared to a simple lexical prefix. A related idea is the use of external lexical signals: GlossBERT reformulates word sense disambiguation as context-gloss sentence-pair classification and, in its strongest variant, highlights the target word in the context with quotation marks [13]; unlike the prefix studied here, it relies on external WordNet glosses rather than the target lemma alone.

Overall, prior work motivates our study but does not isolate the contribution of span highlighting and lexical prefixing under a controlled ablation protocol. Table 1 summarises this positioning against representative prior work. To the best of our knowledge, no prior work jointly isolates span highlighting and a lexical prefix in a 2×2 factorial ablation on SuperGLUE WiC, nor decomposes

Table 1. Representative prior work. Mark. = target/span marking; Pref. = lexical prefix; 2×2 = controlled factorial ablation.

Work	Task	Method	Mark.	Pref.	2×2	Difference from our work
WiC [1]	WiC dataset	baselines	No	No	No	Defines the task; does not study input formatting.
SuperGLUE [2]	benchmark	multi-task	No	No	No	Includes WiC; not methodological.
SenseBERT [7]	WiC	supersense pre-training	No	No	No	Modifies pre-training; ours is input-side only.
Matching the Blanks [8]	relation extr.	entity markers	Yes	No	No	Marker usefulness in RE; we test on WiC.
Zhou & Chen [12]	relation extr.	marker variants	Yes	No	No	Compares punctuation vs special-token markers in RE; we test marking vs a prefix on WiC.
MarkBERT [17]	word seg. / NER	boundary markers	Yes	No	No	In-vocabulary markers in Chinese; help or hurt by realisation; we test marking vs a prefix on WiC.
XL-LEXEME [9]	WiC / LSCD	SBERT + target highlight	Yes	No	No	Bi-encoder; does not ablate marking; no 2×2.
MCL-WiC [10, 11]	MCL-WiC	signal on target word	Yes	No	No	Multilingual; we isolate it against the prefix.

the prefix into lemma and POS components. We do not claim to introduce target marking, nor that newly initialized markers can underperform—an effect already documented in relation extraction [12]. Our contribution is to establish this effect, under statistical control, in the WiC cross-encoder setting, and to contrast marking against a lexical prefix.

3 Methodology

3.1 Task and Data

A WiC example is a quadruple (s_1, s_2, w, y) , where s_1 and s_2 are two sentences, w is the target word present in both, and $y \in \{0, 1\}$ indicates same-sense usage. The model learns $f(s_1, s_2, w) \rightarrow y$. The dataset provides w in its base form (the

word field), which we refer to as the target lemma throughout. When fine-tuning a cross-encoder, the two sentences are concatenated with separator tokens, but w is not explicitly privileged. We study two input-augmentation strategies:

- **Span highlighting** (a positional signal): the target occurrence in each sentence is wrapped with the special tokens `<TW>` and `</TW>`. These tokens are added to the tokenizer and their embeddings are learned from scratch during fine-tuning; they do not belong to the model’s pre-training vocabulary.
- **Lexical prefix** (a lexical anchor): the target word (the dataset `word` field) is prepended to each sentence in the form `word (POS):`. This signal reuses in-vocabulary tokens and provides an explicit lexical cue without introducing new symbols.

The two signals are not equivalent: markers introduce new tokens, whereas the prefix reuses in-vocabulary tokens.

We use the SuperGLUE version of WiC: 5,428 training, 638 development, and 1,400 test examples with hidden labels. At the time of writing, we could not obtain an official test-set score because no active official submission route was available to us; all results are on labelled data, and our contribution is a within-backbone comparison under an identical protocol.

3.2 Input Conditions and Prefix Decomposition

Table 2 shows the six input conditions. The four core conditions form the 2×2 factorial design; two additional conditions decompose the prefix. For the conditions with span highlighting, the tokens `<TW>` and `</TW>` are added to the tokenizer as special tokens and the embedding matrix is resized: the marker vectors are therefore initialized from scratch and learned entirely during fine-tuning.

The prefix has the surface form `word (POS):`. We verified the availability of the POS attribute directly in the data: the official SuperGLUE WiC release does not include a part-of-speech field: across all 7,466 records (training, development, and test) the records carry no `pos` key, so every example in our labelled pool resolves to the default tag *noun*. Consequently, the POS component of the prefix is a constant token that carries no lexical-class information, and the prefix is effectively lexical—it anchors the target lemma. The two decomposition conditions `word_only` and `pos_only` make this explicit: the former prepends only the lemma (`word:`); the latter prepends only the constant POS tag (`(noun):`), and thus acts as a negative control.

3.3 Models and Training

We evaluate `microsoft/deberta-large` [5] and `roberta-large` [4]. The hyperparameters (Table 3) were tuned once on DeBERTa-large with Optuna [15] and then kept frozen across all conditions and both backbones; within-backbone comparisons are fully comparable, while absolute cross-backbone comparisons must be interpreted with caution since RoBERTa-large was not tuned separately. No

Table 2. Six input conditions: four core (2×2 design) and two decomposition conditions.

Condition	Span	Prefix type	Example
full	Yes	word + POS	place (noun): ...my <TW>place</TW> later?
no_highlight	No	word + POS	place (noun): ...my place later?
no_prefix	Yes	none	...my <TW>place</TW> later?
plain	No	none	...my place later?
<i>Prefix decomposition conditions:</i>			
word_only	No	word only	place: ...my place later?
pos_only	No	POS only	(noun): ...my place later?

Table 3. Main training configuration, kept frozen across all conditions and backbones.

Parameter	Value
Backbones	microsoft/deberta-large, roberta-large
Max sequence length	128
Epochs	4
Precision	FP32
Effective batch size	16
Learning rate	8.6557×10^{-6}
Weight decay	0.0384
Warmup ratio	0.0760
Hidden dropout	0.0935
Attention dropout	0.0666
Label smoothing	0.0209
Selection metric	accuracy

early stopping is used, to keep the training budget constant across conditions. Training is performed in FP32 for numerical stability with DeBERTa’s disentangled attention mechanism.

3.4 Evaluation Protocol

An initial analysis on the development set ($n = 638$) suggested the trends reported below but had limited statistical power and a smaller effect magnitude (the highlighting effect on DeBERTa-large grows from about 0.3 pp there to 2.26 pp out-of-fold): under per-seed McNemar tests with Holm correction only the prefix effect on DeBERTa-large reached significance. We therefore read absolute magnitudes with caution and rely on the paired contrasts. We therefore adopt as our main analysis a 5-fold cross-validation (CV) on the union of training and development splits ($N = 6066$), which increases statistical power at the cost of comparability with the official splits (see Section 6). A single, fixed stratified cross-validation split¹ is reused across all conditions and backbones, so that the

¹ The split uses a fixed random seed to ensure reproducibility across all runs.

Table 4. Cross-validated OOF accuracy and F1 (5-fold, $N = 6066$), with mean $\pm$ std over folds and delta vs. **full**.

Backbone	Condition	Acc. (%)	F1 (%)	Δ vs. full (pp)
DeBERTa-large	full	78.55 ± 0.76	78.54 ± 0.75	+0.00
	no_highlight	80.81 ± 1.58	80.77 ± 1.60	+2.26
	no_prefix	78.52 ± 1.70	78.48 ± 1.70	-0.03
	plain	79.29 ± 0.83	79.27 ± 0.84	+0.74
	word_only	79.67 ± 1.60	79.63 ± 1.63	+1.12
	pos_only	80.20 ± 1.36	80.16 ± 1.38	+1.65
RoBERTa-large	full	77.37 ± 1.21	77.28 ± 1.22	+0.00
	no_highlight	77.88 ± 1.14	77.83 ± 1.17	+0.51
	no_prefix	76.66 ± 1.01	76.57 ± 0.98	-0.71
	plain	75.70 ± 1.32	75.60 ± 1.34	-1.67
	word_only	77.53 ± 1.40	77.48 ± 1.42	+0.16
	pos_only	75.95 ± 1.19	75.82 ± 1.27	-1.42

out-of-fold (OOF) prediction vectors remain paired and McNemar’s test applies. For each fold, the best checkpoint is selected by held-out-fold accuracy; this is standard practice in cross-validation and introduces at most a small optimistic bias on absolute accuracy, while the within-fold paired contrasts are much less exposed to this bias, since the same selection strategy applies identically to all conditions within each fold. Execution order does not affect results because every run starts from a freshly initialized model with its own seed. Holm–Bonferroni correction is applied to the family of all 7 contrasts per backbone (4 factorial + 3 decomposition), as detailed in Table 5.

4 Experimental Results

4.1 Per-Condition Accuracy

Table 4 reports the 5-fold cross-validated accuracy and F1 for each condition. Two patterns are visible at the descriptive level. First, removing span highlighting does not hurt and on DeBERTa-large clearly helps: **no_highlight** achieves 80.81% OOF accuracy, 2.26 pp above **full**. Second, the prefix is beneficial: on both backbones **no_highlight** (prefix, no markers) is substantially above **plain** (no augmentation). Figure 1 visualises these results.

4.2 Statistical Analysis of the Contrasts

Table 5 reports all 7 contrasts per backbone (4 factorial + 3 decomposition). Holm correction is applied to the family of all 7 contrasts per backbone jointly. Figure 2 gives a forest-plot view of the four core contrasts.

Span highlighting does not improve either backbone and significantly harms DeBERTa-large. On DeBERTa-large, **full** vs. **no_highlight** is -2.26 pp (CI

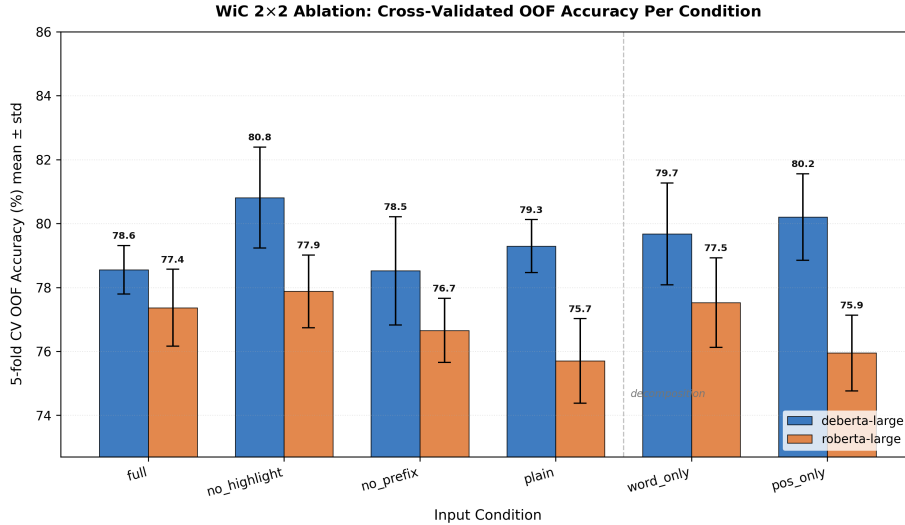


Fig. 1. Cross-validated OOF accuracy per input condition (5-fold, $N = 6066$), mean $\pm$ std over folds. The dashed vertical line separates the four core 2×2 conditions from the two prefix-decomposition conditions.

$[-3.20, -1.30]$, $p_{\text{Holm}} < 0.001$): adding the markers lowers accuracy significantly. On RoBERTa-large the same contrast is -0.51 pp and non-significant. In no setting does highlighting yield a statistically robust improvement.

The lexical prefix significantly improves both backbones when span markers are absent. The **no_highlight** vs. **plain** contrast (prefix added, no markers) is positive and significant on both: $+1.52$ pp (CI $[+0.68, +2.34]$, $p_{\text{Holm}} = 0.003$) on DeBERTa-large and $+2.18$ pp (CI $[+1.17, +3.15]$, $p_{\text{Holm}} < 0.001$) on RoBERTa-large. The pattern of contrasts is also consistent with a negative interaction: the difference between the two highlighting contrasts, (**full** vs. **no_highlight**) minus (**no_prefix** vs. **plain**), is -1.49 pp on DeBERTa-large and -1.47 pp on RoBERTa-large. We report this difference-in-differences descriptively rather than testing it formally; in practice, the prefix is most useful when highlighting is absent.

The lemma, not the constant POS tag, drives the prefix effect on RoBERTa-large. On RoBERTa-large, **word_only** vs. **plain** is $+1.83$ pp ($p_{\text{Holm}} = 0.001$); **pos_only** vs. **plain** is $+0.25$ pp (non-significant); **word_only** vs. **pos_only** is $+1.58$ pp ($p_{\text{Holm}} = 0.006$). Since the POS tag is a constant (*noun*), this is the expected pattern of a negative control: the lemma carries the signal and the constant tag does not. These RoBERTa-large decomposition contrasts are listed in Table 5. On DeBERTa-large no decomposition contrast reaches significance after Holm correction; in particular **pos_only** has a numerically higher delta than **word_only** on that backbone ($+0.91$ vs. $+0.38$ pp), which prevents a definitive

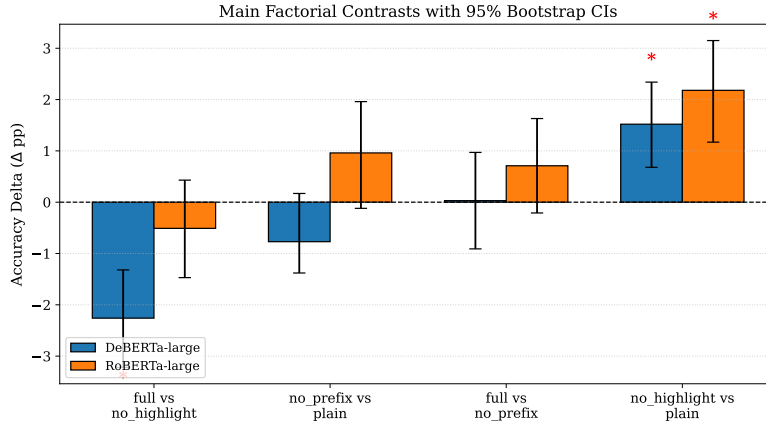


Fig. 2. Main factorial contrasts with 95% bootstrap confidence intervals (5-fold CV, $N = 6066$). For each contrast, $\Delta = \text{acc}(A) - \text{acc}(B)$ in percentage points; negative bars indicate that condition B is more accurate. Asterisks mark contrasts that remain significant after Holm–Bonferroni correction ($p_{\text{Holm}} < 0.05$).

claim. We therefore state the lemma finding as confirmed on RoBERTa-large only.

The 5-fold CV pools the official training and development splits and repartitions them at random; because WiC reuses sentences and target words across examples, absolute OOF accuracy is not directly comparable to the official development or test sets, nor to leaderboard figures such as SenseBERT’s 72.1% test accuracy [7]. We therefore make no claim of leaderboard positioning and rely throughout on the within-fold paired contrasts, which remain valid since any distribution effect applies equally to all conditions on each fold.

4.3 Error Analysis

To understand *how* span highlighting changes predictions, rather than only its effect on aggregate accuracy, we examine the primary contrast on DeBERTa-large, **full** vs. **no_highlight**, at the level of individual out-of-fold predictions. Table 6 reports the agreement between the two conditions over the $N = 6066$ pooled examples.

Removing the markers recovers 504 examples while losing 367, a net gain of 137 correct predictions. This net difference corresponds exactly to the +2.26 pp out-of-fold gap reported in Section 4.2 (Table 5): the headline effect is the residual of two sizable, partially offsetting flows rather than a uniform shift.

We then asked whether the recovered and lost cases differ along simple, interpretable dimensions. We binned the relative position of the target word in each sentence as early, middle, or late, and flagged examples in which the target form occurs more than once. Neither feature separates the two groups: the position distributions are close (for instance, recovered cases have an early-position

Table 5. Cross-validated contrasts (5-fold, $N = 6066$). Holm correction is applied to all 7 contrasts per backbone jointly. A checkmark in the Sig. column marks $p_{\text{Holm}} < 0.05$.

Backbone	Contrast	Δ Acc. (pp)	95% CI (pp)	p_{Holm}	Sig.
DeBERTa-large	full vs. no_highlight	-2.26	[-3.20, -1.30]	< 0.001	✓
	no_prefix vs. plain	-0.77	[-1.71, +0.16]	0.456	
	full vs. no_prefix	+0.03	[-0.91, +0.97]	0.972	
	no_highlight vs. plain	+1.52	[+0.68, +2.34]	0.003	✓
	word_only vs. plain	+0.38	[-0.46, +1.24]	0.808	
	pos_only vs. plain	+0.91	[+0.10, +1.71]	0.159	
	word_only vs. pos_only	-0.53	[-1.35, +0.33]	0.706	
RoBERTa-large	full vs. no_highlight	-0.51	[-1.45, +0.45]	0.602	
	no_prefix vs. plain	+0.96	[-0.12, +1.96]	0.308	
	full vs. no_prefix	+0.71	[-0.21, +1.63]	0.419	
	no_highlight vs. plain	+2.18	[+1.17, +3.15]	< 0.001	✓
	word_only vs. plain	+1.83	[+0.86, +2.77]	0.001	✓
	pos_only vs. plain	+0.25	[-0.66, +1.17]	0.619	
	word_only vs. pos_only	+1.58	[+0.61, +2.51]	0.006	✓

Table 6. Prediction agreement between **full** and **no_highlight** on DeBERTa-large (5-fold OOF, $N = 6066$). *Recovered* examples are classified correctly by **no_highlight** but not by **full**; *lost* examples are the opposite.

	no_highlight correct	no_highlight wrong
full correct	4398	367 (lost)
full wrong	504 (recovered)	797

target in the first sentence 56.0% of the time against 50.1% for lost cases, with no comparable gap in the second sentence), and target repetition is negligible in both groups (below 1%). The effect of the markers is therefore diffuse rather than concentrated on an identifiable subset of examples.

Qualitatively, the discordant cases are ordinary polysemy pairs in both directions. A representative recovered example is *Roll out the paper.* vs. *The drawer rolled open.* (gold label: different senses), which **no_highlight** separates correctly and **full** does not; lost cases are of the same nature in the opposite direction. We did not identify a consistent linguistic property distinguishing when the markers help from when they hurt. This absence is itself informative: it is consistent with the special tokens perturbing the representation broadly rather than interfering with a specific phenomenon.

5 Discussion

The results challenge the intuitive assumption that explicitly marking the target word improves WiC: span highlighting yields no benefit on either backbone and significantly degrades DeBERTa-large. A plausible explanation concerns the

nature of the signals. Span highlighting introduces special tokens absent from pre-training, initialized from scratch and learned on a small dataset; the lexical prefix instead reuses in-vocabulary tokens and provides an explicit lexical anchor. This reading is consistent with evidence from relation extraction, where Zhou and Chen [12] report that markers built from in-vocabulary punctuation can match or outperform markers that introduce new special tokens. Our results extend this observation to the WiC cross-encoder setting and quantify it with statistical control.

The decomposition sharpens this picture on RoBERTa-large. Because we verified that the SuperGLUE WiC release exposes no POS field, the `pos_only` condition is a constant prefix and functions as a negative control: the fact that it does not improve over `plain`, while `word_only` does and significantly outperforms `pos_only`, confirms that the benefit of the prefix is lexical rather than an artifact of prepending any token. This aligns with the distributional intuition behind WiC: the relevant signal is the lemma in context, and a lexical anchor reinforces it. On DeBERTa-large the decomposition is inconclusive—`pos_only` numerically exceeds `word_only`, though neither contrast is significant—consistent with the encoder benefiting from any short prefix as a position cue; separating this from a genuine part-of-speech effect would require the populated POS field that the SuperGLUE release lacks. It is also a non-trivial finding given that WiC is known to carry a strong context bias [16], under which a target-side cue is not guaranteed to help. The negative interaction noted in Section 4.2 further reinforces the picture: the prefix helps most when highlighting is absent, suggesting that the markers can interfere with the lexical anchor.

The error analysis in Section 4.3 both sharpens and bounds this interpretation. The 2.26 pp out-of-fold advantage of `no_highlight` is not a uniform improvement but the net of two sizable, partially offsetting flows (Section 4.3). Crucially, the recovered and lost subsets are not separable by target position or by repetition of the target form, and we could not isolate a linguistic regime in which the markers are systematically harmful—consistent with the out-of-vocabulary tokens `<TW>` and `</TW>`, learned from scratch on a small dataset, broadly perturbing DeBERTa-large’s disentangled content–position attention. Because the change is diffuse and partially self-cancelling, we characterise span highlighting as a net liability for this backbone rather than a targeted failure mode.

This interpretation is deliberately cautious: we have not analysed attention maps or conducted causal interventions, and a feature-attribution analysis (e.g., SHAP) of which input regions most influence the prediction would complement the accuracy-based evidence reported here.

For practitioners, the findings give a concrete guideline. On WiC, span markers initialized from scratch are not worth adding—they bring no gain on either backbone and a significant penalty on the stronger one, DeBERTa-large—whereas a lexical prefix built from the target lemma helps both backbones while reusing tokens the model already understands. Where explicit marking is preferred, an in-vocabulary realization is safer than new special tokens [12]. More

broadly, the value of a target-side signal is architecture-dependent and should be measured, not assumed, rather than treated as a safe default.

6 Limitations

No official test-set score was obtainable, as no active official submission route was available to us at the time of writing. The hyperparameters were tuned on DeBERTa-large and transferred to RoBERTa-large; within-backbone contrasts are valid, but absolute cross-backbone comparisons are not principled. The SuperGLUE WiC release exposes no POS field (verified across all 7,466 records), so we could not assess a genuine lexico-syntactic prefix; the original WiC distribution does encode a noun/verb distinction, and using it—or an external tagger—would be required to test a real POS signal. Relatedly, we did not separate whether the prefix benefit arises from better localization or from easier optimization. Checkpoint selection by held-out-fold accuracy may introduce a small optimistic bias on absolute OOF accuracy, but not on the within-fold paired contrasts. The cross-validation pools the official splits and re-partitions at random, so absolute OOF accuracy is not comparable to the official test set (WiC reuses sentences and target words across examples). Finally, the study covers only two encoder backbones, both used in a cross-encoder configuration; whether the same input-formatting effects hold for bi-encoder architectures—such as the one used by XL-LEXEME [9], where each context is encoded independently—remains an open question.

7 Conclusion and Future Work

We presented a controlled ablation of span highlighting and a lexical prefix for Word-in-Context, together with a prefix decomposition. Across a 5-fold cross-validation on two large encoders ($N = 6066$), span highlighting provided no benefit and significantly lowered DeBERTa-large accuracy, while the lexical prefix significantly improved both backbones when markers are absent; on RoBERTa-large, the lemma rather than the constant POS tag is the active component. The contribution is methodological: statistically grounded evidence on input-design choices often treated as implementation details.

Several directions remain open: an official test-set submission; re-initializing the `<TW>/</TW>` embeddings from in-vocabulary tokens to test the cold-start hypothesis directly; a genuine lexico-syntactic prefix from an external tagger; and extension to other tasks and to DeBERTa-v3 [6].

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